

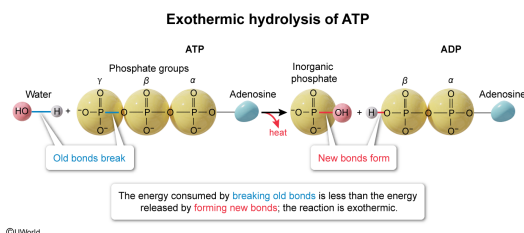
Certain organisms can use ATP hydrolysis ($\Delta H^{\circ'} = -28 \text{ kJ/mol}$) to help maintain their body temperature in cold conditions. Given that cleaving chemical bonds is always endothermic ($\Delta H^{\circ'} > 0 \text{ kJ/mol}$), which of the following explains why ATP hydrolysis is an exothermic reaction?

- ☐ A. The activation energy of hydrolysis is low and provides the energy needed to cleave the phosphoester bond. [8%]
- ✓ ☒ B. More energy is released by the new bonds that form than is consumed by the initial bonds that break during hydrolysis. [65%]
- ☐ C. The adjacent negative charges in the phosphate groups allow bond cleavage to produce thermal energy instead of consuming it. [23%]
- ☐ D. The number of product molecules is greater than the number of reactant molecules. [2%]

Correct

65% answered correctly

Explanation:



Hydrolysis occurs when water breaks a larger molecule into simpler molecules by reacting with one or more bonds in the large molecule (ie, **water adds across bonds**, which **breaks the bonds**). During the hydrolysis process, **two new bonds also form** for each bond that is broken. One new bond forms between a proton from water and one of the atoms from the hydrolyzed (ie, broken) bond, and a second new bond forms between the -OH group from water and the other atom from the hydrolyzed bond.

ATP hydrolysis occurs when **water is added across the bond linking two phosphate groups** in an ATP molecule, which typically results in the formation of ADP and inorganic phosphate. The overall process of ATP hydrolysis is exothermic ($\Delta H^{\circ'} < 0 \text{ kJ/mol}$) and produces heat. All organisms use ATP hydrolysis to help **drive energetically unfavorable reactions**, and some organisms can use the heat of ATP hydrolysis to help maintain body temperature in cold environments.

It is a common misconception that the heat produced by ATP hydrolysis is due to *cleavage* of the P-O bond between phosphate groups. However, bond cleavage always *consumes* thermal energy (ie, it is **endothermic**, $\Delta H^{\circ'} > 0 \text{ kJ/mol}$), so this cannot be the case. In contrast, bond formation is always **exothermic** ($\Delta H^{\circ'} < 0 \text{ kJ/mol}$).

ATP hydrolysis involves both bond cleavage *and* bond formation (ie, the P-O bond is not **cleaved in isolation**), and the reaction is exothermic because the **energy released** by *forming* the new bonds **exceeds the energy consumed** by breaking the old bonds in both ATP and water.

(Choice A) The activation energy of ATP hydrolysis is relatively *high*, allowing organisms to control how quickly ATP is hydrolyzed by activating or inactivating enzymes. In addition, even low

activation energies require at least some energy *input* and cannot provide the energy needed for phosphoester bond cleavage.

(Choice C) Although the negative charges in ATP repel each other and weaken the bonds between phosphate groups, breaking chemical bonds (even weak ones) *always consumes* thermal energy (ie, bond cleavage is always endothermic).

(Choice D) The relative numbers of product and reactant molecules affect the entropy of a reaction, not the enthalpy. In addition, hydrolysis of ATP has the *same number* of reactant molecules (ie, ATP and water) as product molecules (ie, ADP and inorganic phosphate).

Educational objective:

ATP hydrolysis releases thermal energy (ie, it is exothermic) because the energy released by bond formation exceeds the energy consumed by bond breaking during the reaction.

Time spent:0 sec

Subject:
Biochemistry

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System:
Metabolic Reactions